SEGSEE POWER PROJECT · BOOM PAPERMATCHBOOMV3CONFIG

Supply-noise co-design: compiled results

Compiled 2026-10-07. All experiments in `pdn_work/`. Droop numbers are p99.9 per-cycle die-voltage drop unless marked “worst”.

Bottom line

Three pieces combine. A parallel supply, where an off-chip buck carries DC and a slow-switching FIVR handles transients. A small on-die active bank. One droop predictor in the issue stage that both holds the heaviest instructions and places background work. Together they cut p99.9 droop by **55–62%** against a conventional 140 MHz FIVR. The FIVR runs **4× slower**, system efficiency rises **2–4 points**, and performance cost is **0.04–0.5%** on single-thread BOOM.

Reordering instructions on its own does not reduce droop; it has a hard ceiling of a few percent. The lever is when work issues, decided by a forward prediction from the measured supply.

−55 to −62%
p99.9 droop vs FIVR-only at 140 MHz, same 5 nF total die capacitance

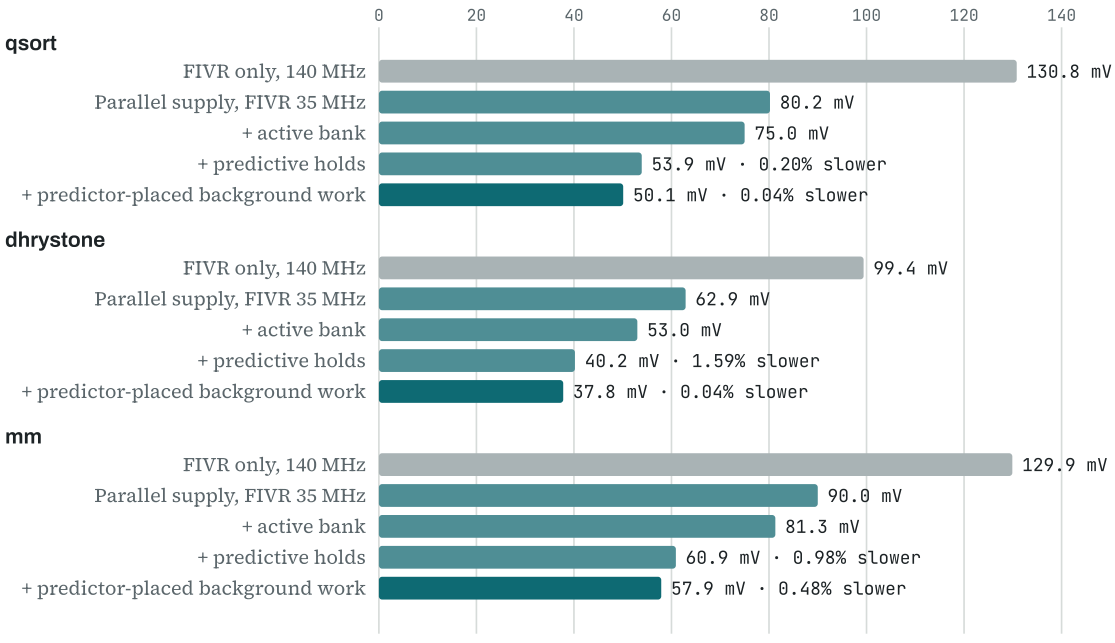
35 MHz
FIVR switching frequency, down from 140, with no droop penalty on the parallel supply

+2.4 to +4.3 pts
system efficiency, 12 V to die (calibrated switching-loss model)

0.04–0.48%
slowdown with predictor-placed background work plus holds

~2×
droop reduction of Grochowski et al. (HPCA'02) at matched GA tuning

How the gain builds up



Single-thread BOOM, full traces, scheduler co-simulated cycle by cycle inside the Pulsim switching circuit. Each bar adds one piece to the one above it. Same 5 nF total die capacitance throughout; the bank takes 0.5 nF from the passive budget. Background work is 1 op/cycle of needed work (for example AES-CTR pads), identical across rows that use it.

CONFIGURATION	QSORT P99.9 / WORST	DHRYSTONE	MM	SLOWDOWN
FIVR only, 140 MHz, normal issue	130.8 / 159.5	99.4 / 143.0	129.9 / 207.9	0
Parallel supply, FIVR 35 MHz (4× L)	80.2 / 134.6	62.9 / 92.5	90.0 / 113.3	0
+ active bank (4.5 + 0.5 nF)	75.0 / 122.6	53.0 / 85.0	81.3 / 110.4	0
+ predictive holds	53.9 / 62.3	40.2 / 51.0	60.9 / 68.8	0.20 / 1.59 / 0.98%
+ predictor-placed background work	**50.1 / 63.8**	**37.8 / 56.2**	**57.9 / 64.4**	**0.04 / 0.04 / 0.48%**

Apparatus

Core and traces

- BOOM `PaperMatchBoomV3Config` RTL (Verilator): 8-wide dispatch, 96-entry IQ, 384 ROB. BACT_CAND trace logs every ready candidate per cycle.
- Replay with real dependencies, the real port map and age priority. Oldest-first reproduces hardware timing exactly (dhrystone 87,312 = 87,312 cycles).
- Power tokens in Chisel (8-bit ROM per uop: 4-bit amplitude, 4-bit duration), verified bit-exact against `curmodel.py`.
- SMT-4/8 model of the Fu et al. ICPP'08 machine (8-wide, unified 96-entry IQ, ICOUNT) built from BOOM trace slices.

Power delivery

- Pulsim (C++ MNA) switching simulation at 50 ps steps, cross-validated against the closed-form model to 1–2%.
- Off-chip 4-phase buck: 500 kHz, per-core share (K = 250) of a 100 A VR, average-current-mode control.
- FIVR: 16 phases, natural-sampled double-edge PWM, type-3 loop at 78 MHz / 43° with 0.6 ns delay (Burton APEC'14 class).
- Package 507 pH / 79.6 mΩ, die 5 nF. Scheduler runs inside the circuit and sees sensed state every cycle.

1 · Supply topology

Direct path only, FIVR only, and both in parallel feeding the same die node. In the parallel case the off-chip loop nulls the FIVR's average current, so the FIVR carries only 1–7% of DC current.

P99.9 DROOP, NORMAL ISSUE	DHRYSTONE	MM	QSORT
Direct path only	163.5	152.5	138.3
FIVR only, 140 MHz	99.4	129.9	130.8
FIVR only, 70 MHz (loop at fsw/1.8)	199.1	271.0	349.4
Parallel, conservative loop (identical at 140 / 70 / 35 MHz)	**63.5**	**89.9**	**78.9**
Parallel, fast loop (crossover ~150 MHz, optimistic)	50.0	66.8	68.6

SYSTEM EFFICIENCY, 12 V TO DIE	DHRYSTONE	MM	QSORT
Direct path only	84.6%	74.0% (anomalous)	85.8%
FIVR only, 140 MHz	82.4%	81.1%	80.1%
Parallel, 35 MHz with 140 MHz inductor	82.5%	82.2%	80.9%
Parallel, 35 MHz with 4× inductor	**84.8%**	**85.4%**	**83.7%**

Conduction losses are simulated from source currents. FIVR switching loss is calibrated at 2.72 pJ per phase-cycle so that FIVR-only at 140 MHz is 90% efficient. The off-chip VR has 5% switching loss and the 1.8 V rail is made at 90%. At 35 MHz the inductor must scale up, or ripple conduction loss cancels the switching savings.

2 · Faithful FIVR re-test

Earlier results used a weak FIVR. Against the Intel-class model (a 100→400 mA step droops 138 mV vs 356 mV with a slow integrator only), several earlier claims did not survive.

CLAIM	WEAK-FIVR RESULT	FAITHFUL-FIVR RESULT	STATUS
Active bank replaces passive decap at the same capacitance	−16 to −34% p99.9	−2 to +4% (no help)	`does not hold`
Reactive holds (REGH) + bank	−56 to −68%	−38 to −48% p99.9, 2.7–6% slowdown	`holds up`
CPU replaces the FIVR fast loop	beat a 140 MHz FIVR at 35 MHz	405–877 mV (holds only), 133–181 mV with dummies	`fails`
Lower FIVR switching frequency via the CPU	4× lower	No bandwidth loss at lower fsw with 16 phases; under the fsw/1.8 rule nothing recovers it	`fails; parallel supply delivers it instead`

3 · Does reordering help?

Tested on single-thread BOOM, on an SMT-4/8 model with Fu-et-al.-level choice, and with a perfect-model predictive selector. Every policy issues the same number of ops as oldest-first.

TEST	RESULT
Single-thread, minimise-bank-need reorder (T)	No droop change (150.8 → 150.5 mV)
Supply-following / spread-spectrum reorder	Mixed; resonance-band energy unchanged (mm 17.8% → 17.9%)
SMT-8, minimise-bank-need reorder	Bank charge −14 to −22% at ≤0.1% runtime change; droop ±3%
SMT-8, state-of-charge-aware selection, half-size bank	Bank shortfall −28 to −76%
SMT, predictive selection with exact PDN model	p99.9 within ±5% (best −5.6%), worst case mixed
Under tuned predictive holds (GA)	Tie with oldest-first (0.1% fitness)

Why there is a ceiling. The top 1% of current steps are mostly issue-*count* rises (+2.2 to +2.4 ops/cycle at ~80 pC each), which a work-conserving selector cannot change. What it can move must still issue within about one resonance period (~10 cycles), so charge per period is nearly conserved. Reordering takes pressure off the bank, but does not reduce droop.

4 · Predictive holds

Each cycle, predict die voltage 13 cycles ahead from the supply's measured impulse response, the current history, the committed tails of in-flight ops, and the candidate picks. If the predicted dip exceeds the target,

hold the heaviest picks, one at a time, until it does not. Target = 0.68 × oldest-first p99.9 (GA-tuned). A starvation guard forces an op through after 19 holds.

SETTING	P99.9	WORST	SLOWDOWN
SMT, RLC model, exact predictor, target 0.7×	-30%	-37 to -47%	-0.7 to +1.3%
Reactive bank-gated holds (TDH) / issue-ramp limit (RAMP)	≈0 to +12%	up to +23%	≈0
GA-tuned ours, realistic HW view, held-out mixes	**-19 to -22%**	**-23 to -32%**	**-0.6 to +0.5%**
GA-tuned Grochowski et al. HPCA'02, same HW view	-2 to -6%	+2 to -11%	≈0
Grochowski at zero latency (ideal), ≤1% slowdown	-8 to -14%	–	≤1%
SMT on parallel supply (Pulsim-identified kernel)	-9 to -16%	-16 to -41%	-0.6 to +0.7%
Same, RLC stand-in predictor (model mismatch)	-7 to -11%	+3 to -15%	≈0
SMT, parallel 35 MHz + bank + holds	**-32 to -43%**	**-44 to -54%**	**≈0**
Single-thread closed loop, parallel 35 MHz + bank + holds	-32 to -36%	-39 to -54%	0.2–1.6%
Single-thread, reactive holds (REGH), same supply	-35 to -43%	-37 to -56%	3.4–9.3%

"Realistic HW view" means 4-bit amplitude × 4-bit duration tokens and the PDN state sensed 2 cycles late. GA: population 48, 20 generations per family, 3 training slices, validated on unseen mixes. The identified kernel matters most for the worst case: the RLC stand-in sometimes makes it worse.

5 · Deferred and useful filler work

The same predictor also places work that must happen anyway: into current dips, and away from predicted droops. Because total work is fixed, the droop gain costs no energy.

CONFIG (SINGLE-THREAD, FINAL SUPPLY)	P99.9	WORST	OVERSHOOT	SLOWDOWN	EXTRA ENERGY
Background work at constant rate (0.5 ops/cycle)	-1 to 0%	-5 to +2%	-11 to -1%	0	needed work
Same work, predictor-placed	**-9 to -13%**	**-8 to -14%**	**-13 to +8%**	**0**	**needed work**
Predictor-placed (1 op/cycle) + holds	**-29 to -33%**	**-34 to -48%**	**-18 to -28%**	**0.04–0.48%**	**needed work**
Classic dummy fill + holds	-23 to -31%	-24 to -48%	-25 to -41%	0.13–0.84%	6.5–10.8% wasted

Redundant execution (DMR) as scheduled work

Every integer op is re-executed within a deadline. Overdue checks reserve ports, which models commit blocking. Coverage is 100% in all rows; the checks add 20–34% charge. Percentages are vs the same core without DMR.

DMR SCHEDULING	DHRYSTONE P99.9 / WORST	MM	QSORT	SLOWDOWN
Eager (next free port)	+40% / +51%	+44% / +41%	+37% / +21%	0.1–1.9%
Predictor-placed, 64-cycle deadline	+6% / +24%	-2% / +6%	+10% / -1%	0.1–2.9%
Predictor-placed + holds	**-7% / -24%**	**-22% / -31%**	**-18% / -33%**	**1.2–3.4%**

Full integer DMR, which normally costs about 40% more droop, ends up with less droop than the unprotected core. Opportunistic overshoot-only re-execution was neutral (±3%, 1–5% coverage). Filler current lands in the next trough of the supply's 7–10-cycle ring.

6 · Clock jitter from supply noise

BASELINE → FINAL DESIGN	QSORT	DHRYSTONE	MM
Cycle-to-cycle supply change, p99.9	59.9 → 36.0 mV	44.1 → 37.0 mV	61.8 → 37.5 mV
Worst cycle-to-cycle change	82 → 45 mV	67 → 46 mV	101 → 51 mV
4-cycle supply change, p99.9	194 → 97 mV	139 → 78 mV	196 → 97 mV
Estimated jitter, p99.9 (0.35 ps/mV, ~300 ps tree)	**21.0 → 12.6 ps**	**15.4 → 13.0 ps**	**21.6 → 13.1 ps**

Most of the jitter gain comes from the parallel supply. Holds mainly reduce droop depth, and on dhrystone they slightly raise cycle-to-cycle change (33.2 → 37.0 mV). The ps conversion uses an alpha-power-law sensitivity estimate.

7 · Earlier groundwork

- The HTIF program-load phase was 72–87% of every raw trace and had to be stripped. Real issue contention is 9–47% of active cycles (mean choice set 2–3.5).
- Closed-loop sensed tokens are required; open-loop token models were about 2× optimistic on slowdown.
- Real switched-cap charge pump decomposed per execution unit: CSR needs the most decap despite the lowest mean current.
- Meng & Saleh JSSC 2009 active decap: all five gain-equation examples re-derived exactly.
- Dummy fill (THF 200) alone: −8 to −16% droop for +3–4% energy.

8 · Novelty

ELEMENT	STATUS	CLOSEST PRIOR ART
Droop-triggered issue throttling, PDN-convolution voltage estimate	prior art	Grochowski, Ayers, Tiwari HPCA'02; Intel DE10297598T5
Charge-budget throttling, queue-activity predictors	prior art	IBM US11625087, US10552250, US7937563
Dummy ops, pipeline damping	prior art	Grochowski; Powell & Vijaykumar ISCA'03
Store-on-overshoot / bidirectional buffer bank	prior art	TI US10523114, NXP US12476539, Richtek US7274177, UCF US7251113
Parallel main + fast auxiliary regulator	prior art	NXP US12476539; envelope-tracking switcher + linear
Forward prediction from a measured supply kernel + committed op tails, selective minimal holds	not found	~2× Grochowski at matched tuning
Predictor-placed useful work (encryption pads) as the current-shaping filler	not found	—
Predictor-scheduled DMR checks, reliability without a droop penalty	not found	Relates to Fu et al. ICPP'08 (IQ reliability)
System co-design: parallel supply + bank + one predictor driving holds and filler	not found	—

Self-search of patents and papers read in full; not a legal clearance.

9 · Caveats

- The per-uop current model is approximate (±30–40% absolute). It was rebuilt after the original library was lost. Relative comparisons are the reliable part.

- The parallel supply's result relies on 16-phase double-edge PWM passing control content above fsw. With a stricter modulator it may weaken.
- Switches are ideal. Switching loss is calibrated, not simulated, and PWM chatter at 4× L (~1.6× transitions) is not charged.
- SMT results come from a replay model built on single-thread BOOM traces, not SMT RTL. Single-thread results use real BOOM ports and dependencies.
- The predictor is a 256-tap convolution plus a 13×10 effect matrix. It needs a low-order IIR and a per-token droop-signature ROM to fit the issue-select path; hardware cost is not yet measured.
- Supply is a per-core share on one lumped die node: no multi-core rail and no spatial grid. The 1.8 V rail in the parallel case is ideal.
- Workloads are qsort, dhrystone and mm. There are no SPEC runs and no power-virus stressmarks yet.

10 · Next steps toward a paper

1. Gate-level per-cycle power from BOOM in ASAP7 (Chipyard Hammer); refit the power-token ROM from it.
2. SPEC CPU2017 SimPoints plus di/dt stressmarks, since the guardband is set by adversarial code.
3. Multi-core shared rail and a published PDN profile; validate Pulsim against ngspice.
4. Chisel predictor (IIR plus droop-signature ROM) synthesised for area and timing.
5. Full GA-tuned baseline suite, including adaptive clock stretching, and end-to-end energy/performance at iso-guardband.
6. Sensitivity: kernel drift, token error, latency, bank size, fsw, decap budget.

Files

TOPIC	CODE	DATA
Supply chain model (cases A, B, D), bank, energy	`pdnchain.py`, `t3design.py`	—
Faithful FIVR re-test	`fivr_exp.py`	`fivr_exp.jsonl`
Parallel supply, efficiency	`par_exp.py`, `par_energy.py`	`par_exp.jsonl`, `par_energy.json`
SMT model, reorder, SOC, MPC	`smt_replay.py`, `smt_exp.py`, `soc_exp.py`, `mpc_exp.py`	`smt_exp*.jsonl`, `soc_exp.jsonl`, `mpc_exp.jsonl`
Predictive holds, GA head-to-head	`hold_exp.py`, `ga_exp.py`	`hold_exp.jsonl`, `ga_champions.json`, `ga_validate.jsonl`
Supply kernels, holds on parallel supply	`ident.py`, `ident2.py`, `hold_par.py`, `hold_par2.py`	`kernels.json`, `hold_par*.jsonl`
Single-thread closed loop, filler, DMR, jitter	`replay2.py`, `st_exp.py`, `wb_exp.py`, `dmr_exp.py`, `jitter_exp.py`	`st_exp.jsonl`, `wb_exp.jsonl`, `dmr_exp.jsonl`, `jitter_exp.json`
Prior-art report	`docs/prior-art-report.md`	